## DEPARTMENT OF ELECTRICAL & ELECTRONIC ENGINEERING

# Experiment No. : 4

# Experiment Name: System Response Analysis using MATLAB

## OBJECTIVES

**1.1 Understanding the basic principle to evaluate Time Response of systems.**
**1.2 Using different inputs (Step, Ramp,and Parabolic) evaluate system time response.**
**1.3 Calculate different parameter of system & compare them theoretically & MATLAB**

## PRELAB WORK:

1. **Read this laboratory tutorial carefully before coming to the laboratory class, so that you know what is required.**
2. Try to follow the lectures of EEE 4701.
3. Familiarize yourself with relevant MATLAB functions and codes necessary for this experiment.

**OBJECTIVE 1.1 Understanding the basic principle to evaluate Time Response of systems.**

## Time Response Overview

The time response represents how the state of a dynamic system changes in time when subjected to a particular input. Since the models we have derived consist of differential equations, some integration must be performed in order to determine the time response of the system. For some simple systems, a closed-form analytical solution may be available. However, for most systems, especially nonlinear systems or those subject to complicated input forces, this integration must be carried out numerically. Fortunately, MATLAB provides many useful resources for calculating time responses for many types of inputs, as we shall see in the following sections.

The time response of a linear dynamic system consists of the sum of the transient response which depends on the initial conditions and the steady-state response which depends on the system input. These correspond to the free (homogeneous or zero input) and the forced (inhomogeneous or non-zero input) solutions of the governing differential equations respectively.

## Stability

The transfer function representation is especially useful when analyzing system stability. If all poles of the transfer function (values of s at which the denominator equals zero) have negative real parts, then the system is stable. If any pole has a positive real part, then the system is unstable. If we view the poles on the complex s-plane, then all poles must be in the left half plane (LHP) to ensure stability. If any pair of poles is on the imaginary axis, then the system is marginally stable and the system will oscillate. The poles of a LTI system model can easily be found in MATLAB using the pole command, an example if which is shown below:

```
s = tf('s');
G = 1/(s^2+2*s+5)
pole(G)
```

```
G =       1
  -------------
s^2 + 2 s + 5

Continuous-time transfer function.
```

```
ans =

  -1.0000 + 2.0000i

  -1.0000 - 2.0000i
```

Thus this system is stable since the real parts of the poles are both negative. The stability of a system may also be found from the state-space representation. In fact, the poles of the transfer function are the eigenvalues of the system matrix A. We can use the eig command to calculate the eigenvalues using either the LTI system model directly, eig(G) or the system matrix as shown below.

```
[A,B,C,D] = ssdata(G);

eig(A)
```

```
ans =

  -1.0000 + 2.0000i

  -1.0000 - 2.0000i
```

## System Order

The order of a dynamic system is the order of the highest derivative of its governing differential equation. Equivalently, it is the highest power of s in the denominator of its transfer function. The important properties of first, second, and higher order systems will be reviewed in this section.

## First Order Systems

First order systems are the simplest dynamic systems to analyze. Some common examples include cruise control systems and RC circuits.
The general form of the first order differential equation is as follows

(1) $\dot{y} + ay = bu \quad \text{or} \quad \tau\dot{y} + y = k_{dc}u$

The first order transfer function is

(2) $G(s) = \dfrac{b}{s+a} = \dfrac{k_{dc}}{\tau s + 1}$

**DC Gain**

The DC gain, ,$k_{dc}$the ratio of the magnitude of the steady-state step response to the magnitude of the step input. From the**Final Value Theorem**, for stable transfer functions the DC gain is the value of the transfer function when s=0. For first order systems equal to$k_{dc} = b/a$

**Time Constant**

The time constant $T_c = \tau = 1/a$ is the time it takes for the system to reach 63% of the steady-state value for a step response or to decrease to 37% of the inital value for an impulse response. More generally, it represents the time scale for which the dynamics of the system are significant.

**Poles/Zeros**

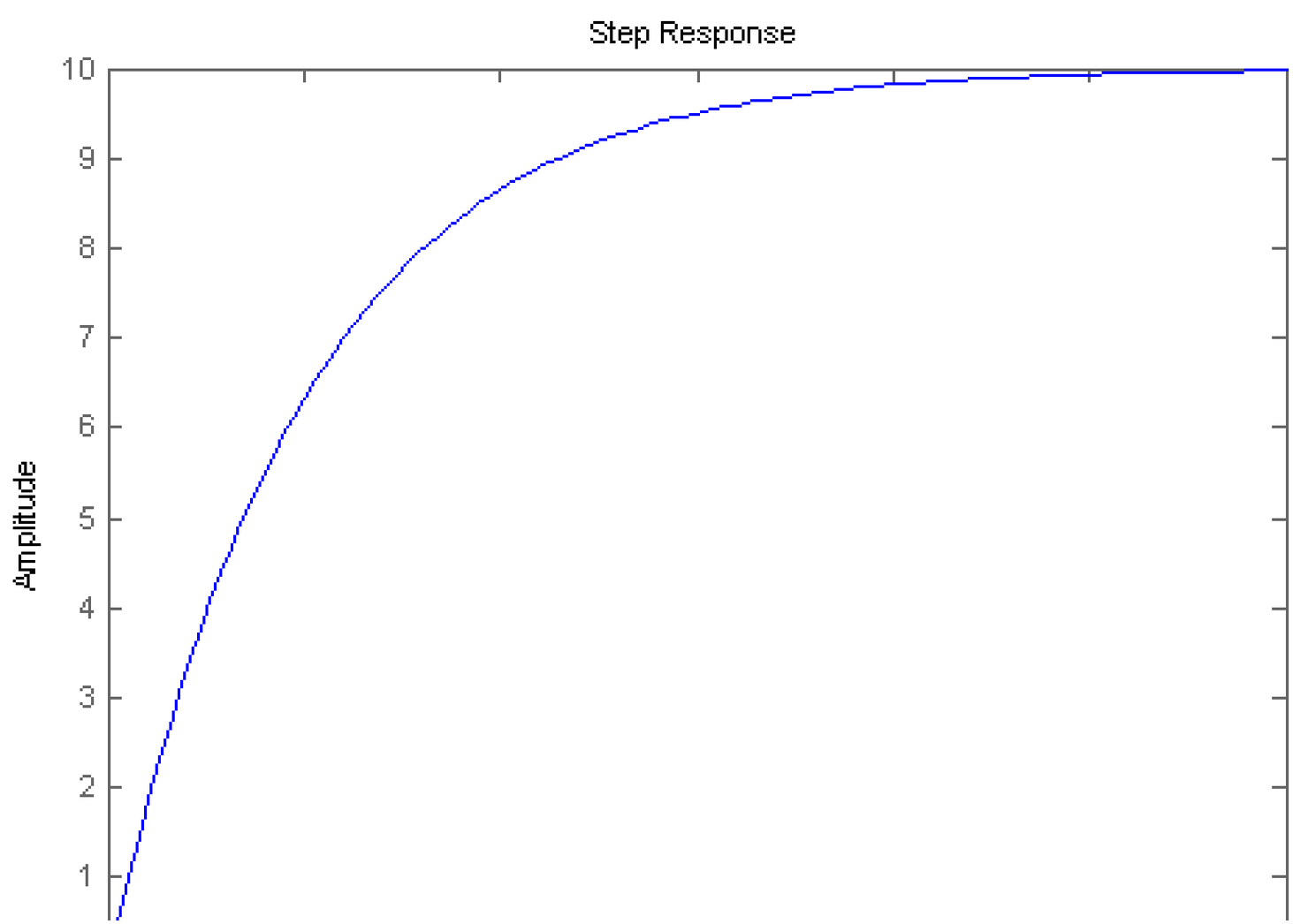
There is a single real pole at $s = -a$. Therefore, the system is stable if $a$ is positive and unstable if $a$ is negative. There are no zeros.

**Step Response**

We can calculate the system time response to a step input of magnitude u using the MATLAB following commands:

```
k_dc = 5;
Tc = 10;
u = 2;
s = tf('s');
G = k_dc/(Tc*s+1)
step(u*G)
```

```
G =    5
  --------
  10 s + 1
Continuous-time transfer function.
```



Note: MATLAB also provides a powerful GUI (LTI Viewer) for analyzing LTI systems which can be accessed using,`ltiview('step',G)`.

If you right click on the step response graph and select **Characteristics**, you can choose to have several system metrics overlaid on the response: peak response, settling time, rise time, and steady-state.

## Settling Time

The settling time, $T_s$, is the time required for the system ouput to fall within a certain percentage (i.e. 2%) of the steady state value for a step input or equivalently to decrease to a certain percentage of the initial value for an impulse input. The settling times for first order system for the most common tolerances are provided in the table below. Note that the tighter the tolerance, the longer the system response takes to settle to within this tolerance, as expected.

| 10% | 5% | 2% | 1% |
|---|---|---|---|
| Ts=2.3/a=2.3Tc | Ts=3/a=3Tc | Ts=3.9/a=3.9Tc | Ts=4.6/a=4.6Tc |

### Rise Time

The rise time, $T_r$, is the time required for the system output to rise from some lower level x% to some higher level y% of the final steady-state value. For first order systems, the typical range is 10%-90%.

**Objective 1.2 Using different inputs (Step, Ramp, and Parabolic) evaluate system time response.**

### Example of Second order system

When the closed-loop system involves a numerator dynamics, the unit-step response curve may exhibit a large overshoot. Obtain the unit-step response of the following system with MATLAB:

$$\frac{C(s)}{R(s)} = \frac{10s + 4}{s^2 + 4s + 4}$$

Obtain also the unit-ramp response with MATLAB.

**Solution.** MATLAB Program 5–21 produces the unit-step response as well as the unit-ramp response of the system. The unit-step response curve and unit-ramp response curve, together with the unit-ramp input, are shown in Figures 5–62(a) and (b), respectively.

Notice that the unit-step response curve exhibits over 215% of overshoot. The unit-ramp response curve leads the input curve. These phenomena occurred because of the presence of a large derivative term in the numerator.

**MATLAB Program 5–21**

```
num = [0  10  4];
den = [1  4  4];
t = 0:0.02:10;
y = step(num,den,t);
plot(t,y)
grid
title('Unit-Step Response')
xlabel('t (sec)')
ylabel('Output')

num1 = [0  0  10  4];
den1 = [1  4  4  0];
y1 = step(num1,den1,t);
plot(t,t,'--',t,y1)
v = [0  10  0  10]; axis(v);
grid
title('Unit-Ramp Response')
xlabel('t (sec)')
ylabel('Unit-Ramp Input and Output')
text(6.1,5.0,'Unit-Ramp Input')
text(3.5,7.1,'Output')
```

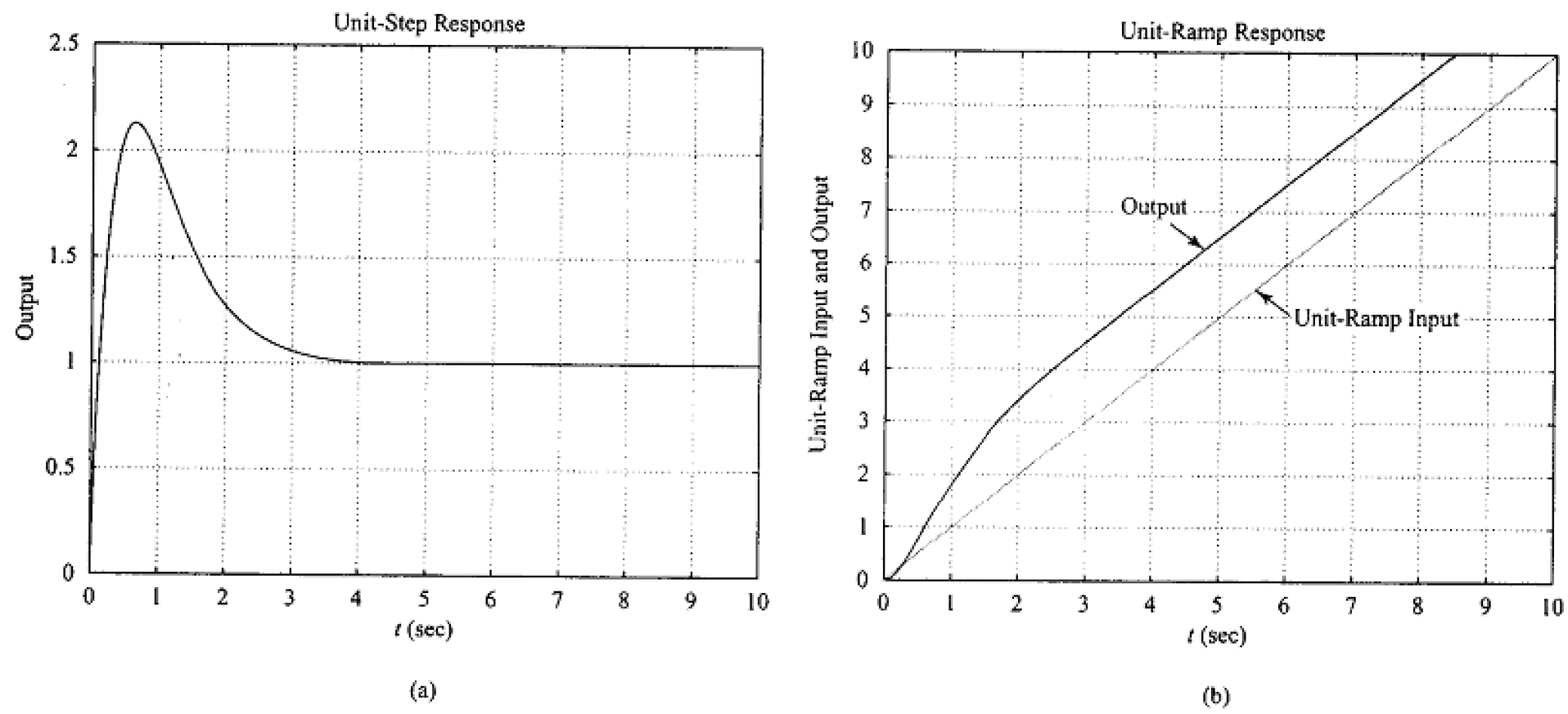



**Figure 5–62**
(a) Unit-step response curve; (b) unit-ramp response curve plotted with unit-ramp input.

## Example of Higher order system

**Solution.** The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{5(s+20)}{s(s+4.59)(s^2+3.41s+16.35)+5(s+20)}$$

$$= \frac{5s+100}{s^4+8s^3+32s^2+80s+100}$$

$$= \frac{5(s+20)}{(s^2+2s+10)(s^2+6s+10)}$$

The unit-step response of this system is then

$$C(s) = \frac{5(s+20)}{s(s^2+2s+10)(s^2+6s+10)}$$

$$= \frac{1}{s} + \frac{\frac{3}{8}(s+1) - \frac{17}{8}}{(s+1)^2+3^2} + \frac{-\frac{11}{8}(s+3) - \frac{13}{8}}{(s+3)^2+1^2}$$

The time response $c(t)$ can be found by taking the inverse Laplace transform of $C(s)$ as follows:

$$c(t) = 1 + \frac{3}{8}e^{-t}\cos 3t - \frac{17}{24}e^{-t}\sin 3t - \frac{11}{8}e^{-3t}\cos t - \frac{13}{8}e^{-3t}\sin t, \qquad \text{for } t \geq 0$$

A MATLAB program to obtain the unit-step response of this system is shown in MATLAB Program 5–20. The resulting unit-step response curve is shown in Figure 5–61.

**MATLAB Program 5–20**

```
% -------------- Unit-step-response --------------

num = [0  0  0  5  100];
den = [1  8  32  80  100];
```

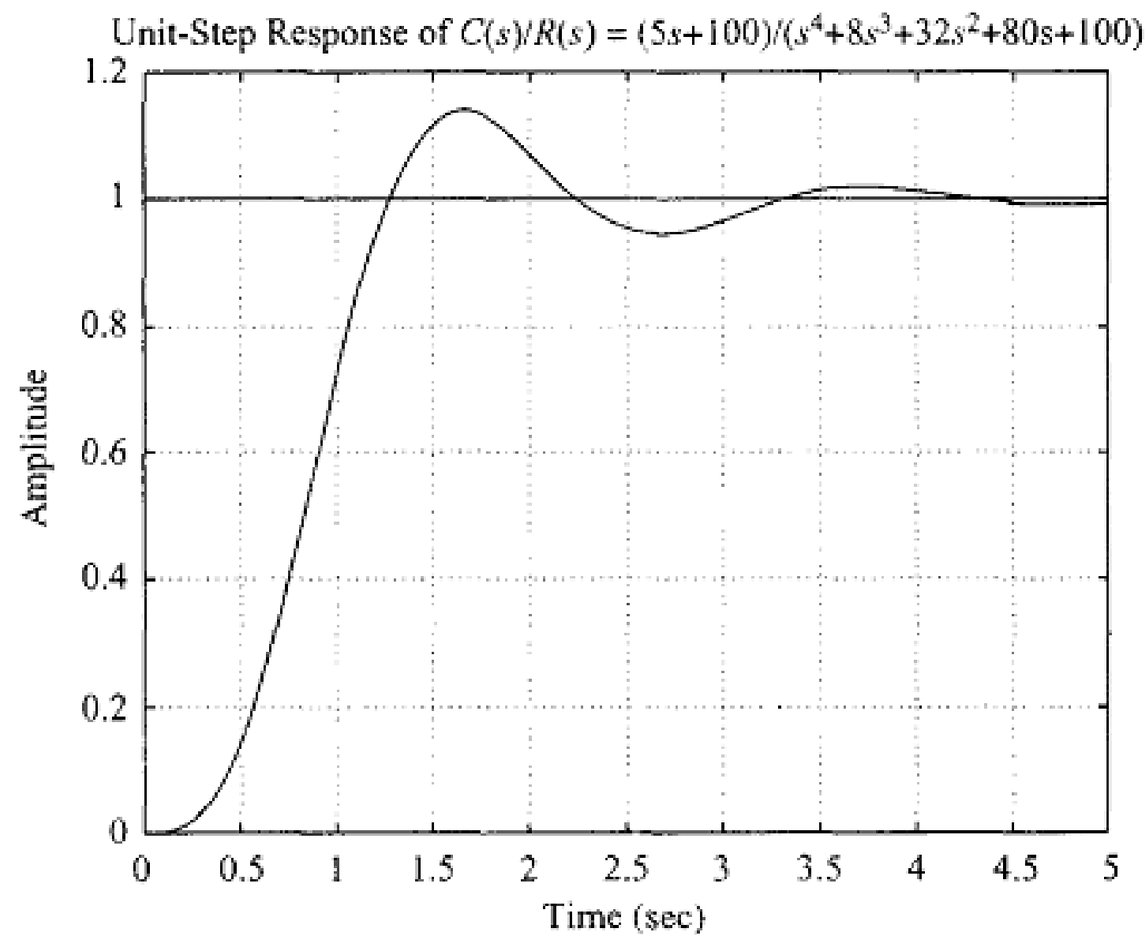



**Figure 5–61**
Unit-step response curve.

**Objective 1.3** **them theoretically & MATLAB**

**Example 6.1:** Consider the following second-order system

$$\frac{Y(s)}{U(s)} = \frac{4}{s^2 + 2s + 4}$$

Using (6.4) and (6.5) we get

$$\omega_n^2 = 4 \Rightarrow \omega_n = 2\,\text{rad/s}, \;\; 2\zeta\omega_n = 2 \Rightarrow \zeta = 0.5$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = \sqrt{3}\ \text{rad/s}$$

The peak time is obtained from (6.15) as

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\sqrt{3}} = 1.82\,\text{s}$$

and the settling time, from (6.20), is found to be

$$t_s \approx \frac{3}{\zeta\omega_n} = 3\,\mathrm{s}$$

The maximum percent overshoot is equal to

$$MPOS = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}100(\%) = 16.3\%$$

The step response of this system obtained by the MATLAB function `[y,x]=step(num,den,t)` with `t=0:0.1:5` is presented in Figure 6.5. It can be seen that the analytically obtained results agree with the results presented in Figure 6.5. From Figure 6.5 we are able to estimate the rise time, which in this case is approximately equal to $t_r \approx 0.8\,\mathrm{s}$.

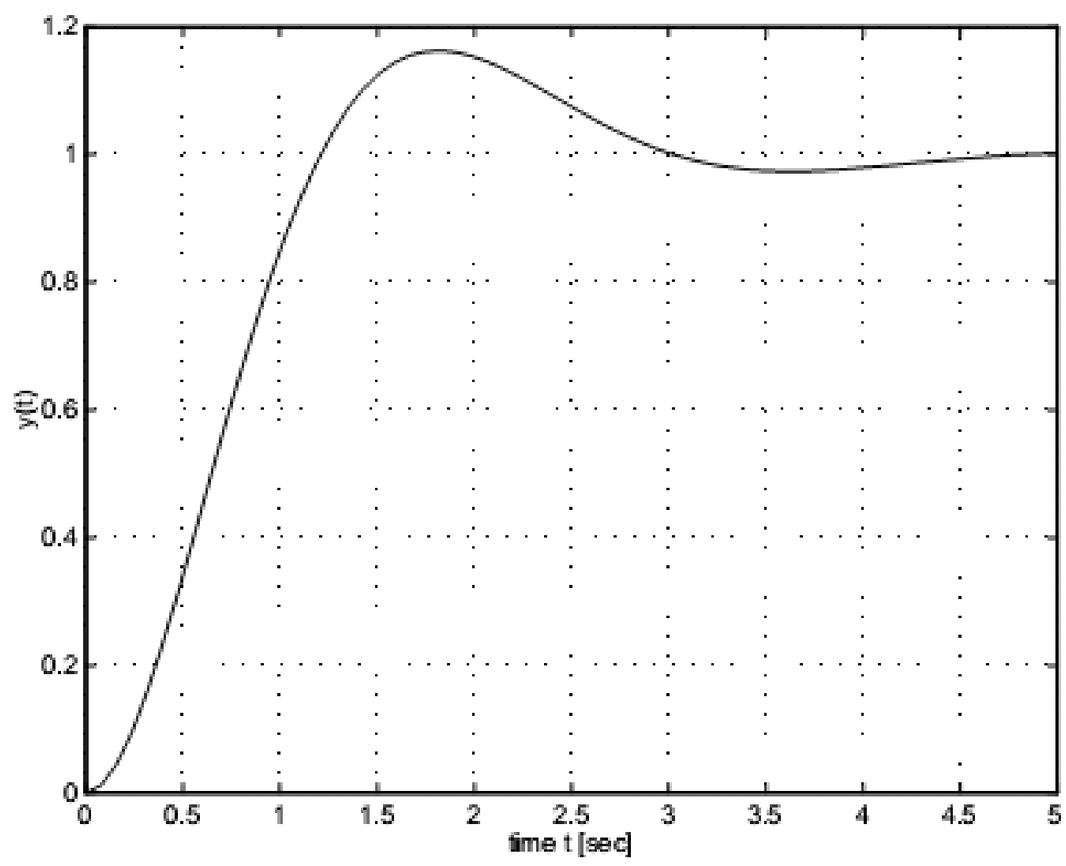



Figure 6.5: System step response for Example 6.1

## Lab Task:

1. Obtain the system response the second order system below

$$\frac{s + ID}{s^2 + 6s + ID}$$

ID means last two digit of matric ID.

2. Obtain both analytical and computational solutions of the unit-step response of a unity-feedback system whose open-loop transfer function is

$$G(s) = \frac{10(s + 15)}{s(s + ID)(s^2 + 5s + ID)}$$

3. Write a MATLAB program to calculate natural frequency, peak time, settling time, rise time of lab task 1

## Report preparation

Please prepare your report with yourself & don't try to copy. Report should cover the following Topics:

1. Objective
2. Apparatus/Utility software:
3. Description of Task:
4. MATLAB code & Result:
5. Analysis of Each output:
6. Discussion: